www.primetechmixer.com

THE RIGHT MIX OF

TECHNOVATION

PRIMETECH

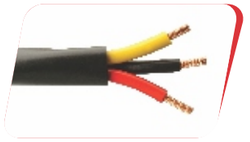
MIXING & SIZE REDUCTION SOLUTIONS WE OFFER FOR THE INDUSTRY



Pipes & Fittings



PVC/WPC Board& Profile



Wire & Cables



Masterbatch



Compounds



Rotomolding

DRIVING FORWARD THROUGH INNOVATION

At Primetech, the era of innovation is renewed everyday. It is through persistent innovation, combined with increasing our efficiency, and technical expertise, that we have arrived at the precipice of being known as the preferred producer and supplier of plastic processing equipment not just in India, but also around the world.

As a company endowed with several recognized accreditations and certification, we are today proudly operating out of a large, scalable facility from the the emerging industrial hub of India, Gujarat. Our industrial campus stands as a testament to our growing desire to predict and satiate market demands with brilliantly designed, efficient products. From a tech-savvy manufacturing floor, to a knowledge-driven research & development wing, we have left no stone unturned to ensure that every equipment and parts from us, exhibits world-class quality and best-in-the-market value.

Driven by the visionary leadership of our management and fueled by the experienced know-how of our entire team, we are continuously growing, learning and adapting to the changing market trends and forever remain committed to helping our clients by delivering them with the equipment that they desire.

PRIMETECH

WE PRIDE IN CALLING OURSELVES P.R.I.M.E

- **P**rincipled with special focus on transparency at all levels
- **R**esourceful with a heart for innovation & the wits for problem solving
- **I**ntegrity & trust-driven business approach backed by promise of quality
- **M**astery of our tradecraft, with sights on futuristic product development
- **E**conomic in every dealing with focus on reducing cost of ownership for clients



THE PRIME VANTAGE

Consistent quality owing to sourcing of component from reputed brands

Better product life, with minimal maintenance and wear and tear

Intelligently built machines, with profound safety measures

Customizable solutions to meet diverse industry & production requirements

Support & service through our network of reputed and authorized professionals

User-friendly & easy to operate machines reducing cost of training

INDUSTRY FOOT PRINT

- India fastest growing brand in Mixing & Size Reduction Equipments
- 30 years of experience
- 3 state of the art manufacturing units spread over an area of 1,00,000 sq ft area equipped with ultra modern and sophisticated machinery
- Offers High Intensity Mixers - Efficient Cooler from 1 kg batch to 500 kg batch capacity
- Serving Different Mixing application of industry including PVC & WPC Dryblend & Materbatches
- Offers Pulverizer from output ranging from 80 kg/ hr to 800 kg / hr.
- Serving Pulverizing application of industry including PVC, cPVC, WPC, Roto-Molding & Materbatches
- Installed capacity to deliver more than 300 units per annum
- 2500+ Mixing Plant successfully commissioned worldwide Including India, Middle East, African Nations, United States & Latin America
- 800+ Pulverizing System successfully commissioned worldwide

HIGH SPEED
HEATER COOLER MIXER with
Bulk Powder Handling &
Conveying Systems



MIXERS



BETTER MIXING

The PRIMETECH- mixer holds a great name for outstanding mixing technology, highest quality standards & competence in field of compounding with its unique machine design & performance.

Primetech high intensity mixer "PHM" is used for all bulk material application like to mix solid, liquid and pasty components where besides friction heat generation with shortest possible cycle time also the way in which the friction is introduced in to the product along with the finest material dispersion is of greater importance.

The unique designs of mixing bowl with large bottom radius and fluidizing blades make PRIMETECH -mixer in to a high performance machine and ideal solution when good quality material are required with high hourly output.

EFFICIENT COOLING

The PRIMETECH cooler mixer "PCM" works effectively in combination with upstream high intensity the PRIMETECH "PHM" for subsequently cooling of hot mixture in order to keep it in free flow condition, suitable to storage transportation and for subsequent processes.

For application in case of P.V.C & various other dry blends where static free, heat stabilized and stress free compounds are suitably required, PRIMETEH cooler mixer "PCM" delivers high heat transfer co-efficient, necessary for obtaining short cooling times for throughput capacities.

Mixing, dispersing process along with de-agglomeration are frequently of equal importance as factor concerning the handling of the machine.



PRODUCT KEY ATTRIBUTE

- All important machine components (mixing vessels, mixing tools, electrical cabinets, etc.) are manufactured in house, thus giving us a very tight control over the quality of our products
- Robust construction
- Only first-class electrical, electronic and pneumatic components, motors and gearboxes of international brands are used in our production
- Ideal mixing Batch timings
- Shorter cooling cycle
- User friendly operation
- Prompt and efficient after-sales service

ALWAYS A STEP AHEAD IN DELIVERING

Innovation | Quality | Performance | Service

MIXING VESSEL AND DEFLECTOR

Internal surfaces in contact with material are made of stainless steel and Double steel jacketed. Contact surfaces are uniquely mirror polished for free flowing of material with deflector influencing the flow of material.





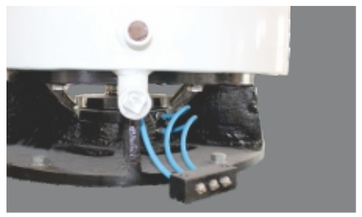
MIXING TOOL

Made from high carbon stainless steel with suitable heat treatment process in order to enhance its wear resistance properties.



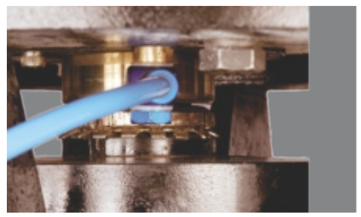
CENTRALIZED LUBRICATION PORT

For easy service & maintenance of the machine



PROVEN PNEUMA-SEAL TECHNOLOGY

Prevents powder infiltration Into bearing and convincingly Increasing the bearing life period.



PERFORMANCE FEATURE

- Heater Mixer with good heating performance
- Suitable for plastic, minerals, chemical and colorant
- Short heating times
- Heat induced by friction
- Homogenizing in minimum of time
- Very robust & easy service construction of mixing vessel & tool
- Twin jacketed mixing vessel
- Fast and complete discharge of mixed material
- Improved tool life
- Long service life and less noise level
- Excellent dispersion and optimum adaptability to each type and size of mixing batch



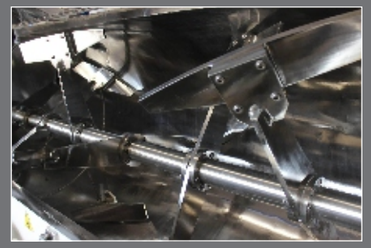
COOLING VESSEL

Large bottom radius with mirror finished contact surface enabling gentle treatment of material due to large surface mixing and low speed.



HORIZONTAL COOLER

Larger Surface area Reduces cooling cycle significantly



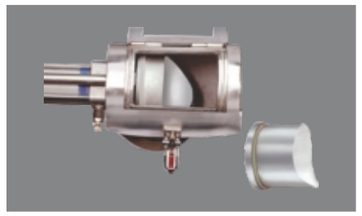
MIXING TOOL

Process enhancing and self cleaning



DISCHARGE VALVE

Piston type valve with complete electro- pneumatic operation for Easy and smooth discharge of a material.



CONTROL PANEL

Panel construction and wiring as per international standards of panel design



PRIMETECH

PRODUCT RANGE

SR. NO.	MODEL	CHARGE BATCH CAPACITY (kg)	MAIN DRIVE PHM (kw)	MAIN DRIVE PCM (kw)	TOOL SPEED PHM / PCM (RPM)	VESSEL CAPACITY PHM / PCM (ltrs)	PRODUCTION CAPACITY (kg/hr)	COOLING SURFACE (sq. m)
1	PHM 10/PCM 40	3	3.7	0.75	910/90	10/40	15	1.3
2	PHM 50/PCM 150	15	7.5	1.5	1000/90	50/150	75	1.7
3	PHM 130 R/PCM 400	40	15	3.7	1000/48	130/400	200	3.5
4	PHM 130/PCM 400	40	18	3.7	1000/48	130/400	200	3.5
5	PHM 200 R/PCM 400	70	30	3.7	1000/48	200/400	350	3.5
6	PHM 200/PCM 400	70	37	3.7	1000/48	200/400	350	3.5
7	PHM 200 R/PCM 600	70	30	5.5	1000/52	200/600	350	5.0
8	PHM 200/PCM 600	70	37	5.5	1000/52	200/600	350	5.0
9	PHM 250 R/PCM 600	90	37	5.5	1000/52	250/600	450	5.0
10	PHM 250/PCM 600	90	45	5.5	1000/52	250/600	450	5.0
11	PHM 350 R/PCM 600	140	55	5.5	1000/52	350/600	700	5.0
12	PHM 350/PCM 600	140	75	5.5	1000/52	350/600	700	5.0
13	PHM 350 R/PCM 1000	140	55	7.5	750/45	350/1000	700	6.5
14	PHM 350/PCM 1000	140	75	7.5	750/45	350/1000	700	6.5
15	PHM 500/PCM 1500	200	110	15	750/65	500/1500	1000	8.0
16	PHM 500/PCM 2000	200	110	18.5	750/65	500/2000	1000	11.0
17	PHM 750/PCM 1500	350	132	15	650/65	750/1500	1500	8.0
18	PHM 750/PCM 2000	350	132	18.5	650/65	750/2000	1500	11.0
19	PHM 1200/PCM 3000	500	250	37	600/65	1200/3000	2500	14.5

*PRODUCTION CAPACITY BASED ON 5 BATCHES PER HOUR.

- Design and Specifications are indicative and as a part of our continuos development we reserve the right to change specifications without any prior notice.
- For customised applications specifications can be altered.
- Output depends on type of material, formulations, processing parameters & ambient temperature.

PERFORMANCE FEATURE

- High heat transfer co-efficient
- Short cooling cycle
- Excellent dispersion with efficient cooling
- Robust, easily serviced and maintained construction
- Corrosion and abrasion resistance
- Easy and perfect cleaning
- Self cleaning, easy fitting and removal
- Intensive cooling of mixed material
- Long life and low noise level
- Twin jacket for cooling with forced circulation of cool water



COMPLETE PLANT AUTOMATION

The PRIMETECH compounding plant automation system works effectively where realisation of high through output rates is combined thought of minimization of labour.

Perfect synchronization of process like material storage, conveying, weighing & mixing are of utmost importance when dealing with complete plant automation solution.

PRIMETECH automated plant system outperforms in all the manner whether process like automatic material handling or complete controlled process sequence are Concerned ,through highly advanced logic based controllers.





STORAGE

Ground hopper for dumping of raw materials with suitable non-dusting system arrangement.

CONVEYING

Mechanical spiral /vacuum conveying.

WEIGHING

Precise and accurate weighing of raw materials and then feeding to heater mixer vessel in exact proportion as per recipe formulation data stored in the system.

MIXING

Guaranteeing complete reproducible and consistent dry blend properties and thus minimizing the extrusion scrap rate.

CONTROL

- Entire process sequence controlled through advanced PLC & HMI for setting and changing the process as well the mixing parameters easily.
- Visual program display sequence by means of luminous signals.

PERFORMANCE FEATURE

- One-switch operation system
- High through output
- Reduction in labour and energy
- 100% "Process in control" through intelligent feedback control system
- Easy operation & maintenance

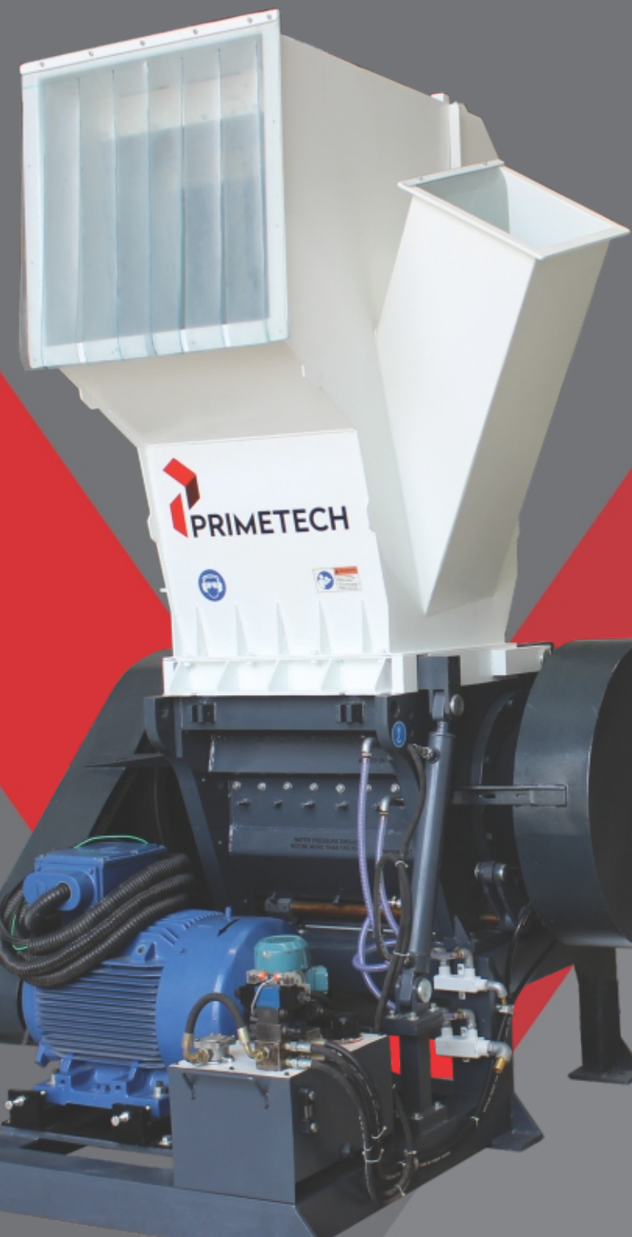
PRIMETECH

Your Reliable Partner for
PLASTIC
SIZE REDUCTION &
REUTILIZATION





GRANULATORS





THE POWERFUL & EFFICIENT GRANULATORS FOR MULTIPLE APPLICATIONS

The PRIMETECH GRANULATORS are comprehensively designed and manufactured for extruded and general recycling needs. The range and versatility of evacuation systems offered make this a popular model for pipes, profile and general recycling applications.

A variety of pipes & profile feed chute and in feed hopper designs are available and are standard.

Large rotor in feed area Granulator for in- and off-line grinding of extruded pieces and general scraps. Grinding chamber with traditional inclined cut allows for intake of heavy and trimmed scraps. Grinding chamber in anti-wear steel to 746 Vickers Rotary blades are standard.

Manual feeding of sheets, pipes, boards, profiles of 6 m long and sprues, runners, preforms, bottles & jars at height of 1.5 m from ground is available as standard.

Grinding chamber with 3 or 5 rotary blades, for processing all resins PVC, WPC, PE, PP, PLA, ABS, EPS, PET and various other engineering plastics.

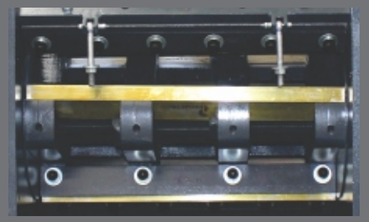
PRIMETECH

ENERGY EFFICIENCY

Lowest running cost expressed in kWh/kg available on the market. Reactive power cost reduced to minimum thus having a direct impact on hourly energy cost.

- Transmission design gives higher inertia: only 20% of installed power consumed during non-productive operation; 40-60% during grinding time

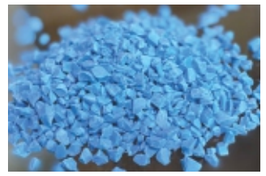




True scissor cut design reduces energy to slice plastic waste hence significantly reduces energy consumption



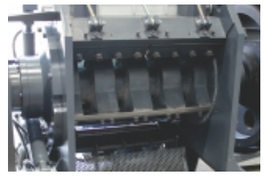
Electro-Hydraulic Hood Opening



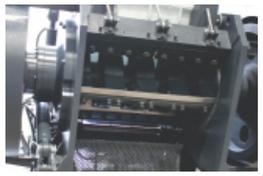
Highest Quality Regrind Constant Cutting Circle Produces Granules A Unifrom Size with Minimum Dust Content Even with Toughest Material



Water Cooling System



Easy Access to Cutting Chamber Importantly without Tools



Electro-Hydraulic Screen Holder Opening



3 knives and 5 knives Rotor Configuration



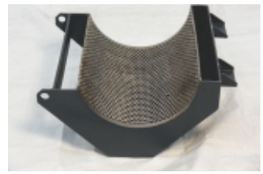
State of Art Electric Control Panel



Cradle Type Screen Holder



Flywheel for High Inertia



Screen Area equal to 50% of rotor Circumference Supports Faster Evacuation of material from Cutting Chamber

HIGH OUTPUT CAPACITY

Stable Performance

High Speed Precision

PULVERIZERS

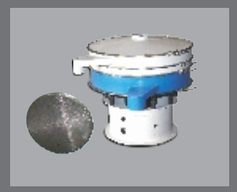
The Primetech PP-series of disc pulverizers are high speed, precision grinders for the processing of medium hard, impact resistant and brittle materials.

Typical applications are the pulverizing of granules to Powder for Roto Moulding, MasterBatch - Compounding, Wood, PVC, WPC Industry.

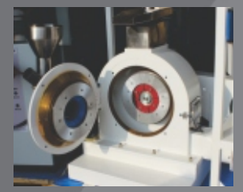
The material to be pulverized is introduced through the center of a vertical fixed grinding disc which is mounted concentrically with an identical high speed rotating disc. Centrifugal force carries the material through the grinding area and the resultant powder is collected with a blower and cyclone system. The Primetech PP series have several technical advantages which ensure ideal performance and operation.

PRIMETECH





Narrow Particle Size Distribution For Better Product Quality



Patent Disc Cutting Teeth Geometry Design



The most **ENERGY EFFICIENT MACHINE**

Key Attributes

- Low power consumption with high throughput
- Economical production cost with consistence output quality
- Low maintenance
- Minimum man power requirement, Pulverizing discs made form German imported high graded alloy steel
- Cooling provision for Stationery disc, Mill body, Duct line, Bearing, Housing
- Monitoring of Main motor load & control system
- Monitoring of Powder Temperature & control system (optional)
- Microprocessor based machine control panel with digital display (optional)
- Exact and easy disc gap adjustment
- Housing design for optimized water-cooling
- Easy to clean
- Broad application field
- Economical and cost effective
- Robust in design
- Quality of Powder according to Specifications

PRIMETECH

Your Reliable Partner for **PLASTIC SIZE REDUCTION** & **REUTILIZATION**

We Offer

- One-Piece pulverizing disc made of German imported high graded alloy steel
- Superior pulverizing disc Surface treatment to boost up it is wear life
- Advance Bearing technology
- Easy adjustments of disc gap
- Unique Design of Housing
- High efficient, Blower & vibro-screen
- Powerful Dual Cooling With Water & Air

Your Advantage

- Pulverizing without using liquid nitrogen
- Excellent flow rate and mixing possibility of the powder
- Compact construction, minimum space requirement
- Most Economical and cost effective
- Narrow particle size distribution with low fines content
- Automatic Load and Temperature controlled.
- Fully automatic continuous operation.
- Safety interlock system

Applications

Lldpe Powder for Rotomoulding,

Recycling of Extrusion Scrap Waste

PVC Pipes & Profile | CPVC Pipes | WPC Boards & Profile Extrusion Scrap | PU | UHMWPE.

Masterbatch, engineering plastics application and various thermoplastic material.



Pipes & profile industry



Powder for Roto Moulding Industry

Output Range

Product Range	PP300	PP400	PP600
Motor Capacity (kw)	15	30	55
Output Capacity	120-200	200 - 300	400-600



PRIMETECH

A total **PULVERIZING SOLUTION** for **MASTERBATCH INDUSTRY**

Output Range

Product Range	PP2030	PP3030	PP5040
Motor Capacity (kw)	15	22	37
Output Capacity	100-125	150-200	350-400



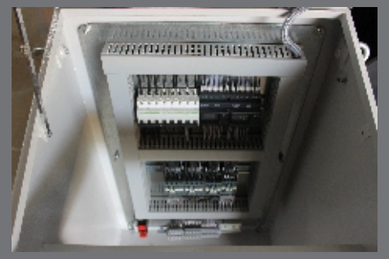




SHREDDERS



Technical Specifications



Intuitive operation

thanks to PLC control

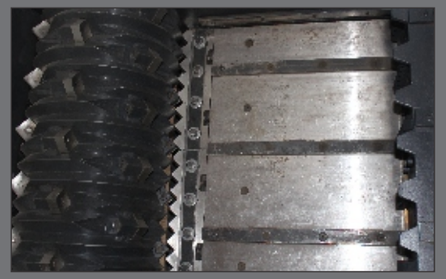
To ensure that the electronics are optimally matched to the machine, we design, build and wire our control cabinets ourselves. Intuitive touch interfaces guarantee quick adjustments. Functions such as setting the ram cycle or stop ensure a high throughput. The built-in overload protection also prevents defects in the machine.



Precise cut with high throughput

with profiled V rotor

The V rotor, specially developed by PRIMETECH, can be used universally and is made of solid material. Its aggressive material feed with up to two rows of knives guarantees high throughput with low power requirements. It can be equipped with hardened steel cutting knives in edge lengths of 30 mm and 40 mm. These can be turned over several times in case of wear, which drastically reduces maintenance costs.



Safe material feed

with load-controlled ram

The ram, which moves horizontally back and forth via hydraulics, feeds the material to the rotor. With PRIMETECH, it can be controlled or cycled automatically depending on the load. If required, of course, it can also be controlled manually.



No material bridges

due to free-cutting hopper design

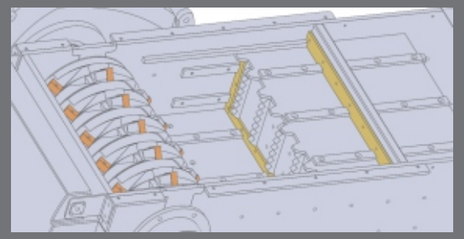
The hopper fulfils several tasks at once. First, the material to be shredded is fed through it – manually, mechanically, or by conveyor belt. The generous opening makes it easy to fill even very large parts. A decisive factor for shredding, however, is its special design, which is rounded at the front and thus effectively prevents material bridging. Even with bulky parts, the shredder cuts itself free. If required, hopper extensions and lids with gas pressure springs are available.



Durable rotor bearings

thanks to offset design

A rotor bearing must be robustly designed due to the extreme stresses it is subjected to while it's in operation. To prevent dust or foreign matter from getting inside it, we have set it apart from the machine frame. This also makes it very maintenance-friendly and easy to access.



Safe material feed

with load-controlled ram

The ram, which moves horizontally back and forth via hydraulics, feeds the material to the rotor. It can be controlled or cycled automatically depending on the load. If required, of course, it can also be controlled manually. For even more aggressive feeding.



Homogeneous shredding results

thanks to flexible, interchangeable screen

Adapt the screen to your needs. The smaller the hole diameter, the finer the shredded material that is discharged. Screens can be exchanged flexibly and are bolted as standard. A screen basket that opens downward ensures even faster screen changes and simplified maintenance.



Efficiently absorb vibrations

with stable rubber feet

Vibration-damping machine feet ensure a secure footing and help to significantly reduce disruptive vibrations in the surrounding area. Since the machine does not have to be anchored to the ground first, installation is particularly flexible and convenient.

Optimally protected hydraulics

installed in the machine frame



CLEAN DISCHARGE OF MATERIAL

by suction, screw or conveyor belt

Depending on your needs and frame design (elevated machine with conveyor belt cutout or side spout), you have the option of discharging shredded material either by air suction, discharge screw or classic conveyor belt. we are also experts in conveyor technology and can supply you with a turnkey solution from a single source

Applications



Powerful drive

with gearboxes



PRIMETECH

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